

The Cellular Concept

- Outline
 - Definitions
 - Frequency Reuse
 - Channel assignment strategies
 - Handoff strategies
 - Interference and system capacity
 - Trunking and grade of service
- Book: Wireless Communications, Rappaport (Chapter-2).

Simplex, half duplex, full duplex

Mobile radio transmission systems may be classified as *simplex*, *half-duplex* or *full-duplex*. In simplex systems, communication is possible in only one direction. Paging systems, in which messages are received but not acknowledged, are simplex systems. Half-duplex radio systems allow two-way communication, but use the same radio channel for both transmission and reception. This means that at any given time, a user can only transmit or receive information.

Full duplex systems, on the other hand, allow simultaneous radio transmission and reception between a subscriber and a base station by providing two simultaneous but separate channels (frequency division duplex or FDD) or adjacent time slots on a single radio channel (time division duplex, or TDD) for communication to and from the user.

Frequency division duplexing (FDD) provides simultaneous radio transmission channels for the subscriber and the base station, so that they both may constantly transmit while simultaneously receiving signals from one another. At the base station, separate transmit and receive antennas are used to accommodate the two separate channels.

In FDD, a pair of simplex channels with a fixed and known frequency separation is used to define a specific radio channel in the system. The channel used to convey traffic to the mobile user from a base station is called the *forward channel*, while the channel used to carry traffic from the mobile user to a base station is called the *reverse channel*.

Time division duplexing (TDD) uses the fact that it is possible to share a single radio channel in time, so that a portion of the time is used to transmit from the base station to the mobile, and the remaining time is used to transmit from the mobile to the base station.

- Basic cellular system consists of
 - Mobile stations (e.g. mobile phones) (MS)
 - users transceiver terminal (handset, mobile)
 - Base stations (BS)
 - fixed transmitter usually at centre of cell
 - includes an antenna, a controller, and a number of receivers
 - Mobile switching center (MSC)
 - Sometimes called a mobile telephone switching office (MTSO)
 - handles routing of calls in a service area
 - tracks user
 - connects to base stations and PSTN

1G Mobile Phone



Dr. Martin Cooper of Motorola, made the first US analogue mobile phone call on a larger prototype model in 1973. This is a reenactment (tekrarlamak) in 2007

Mobile Switching Center (MSC Server)



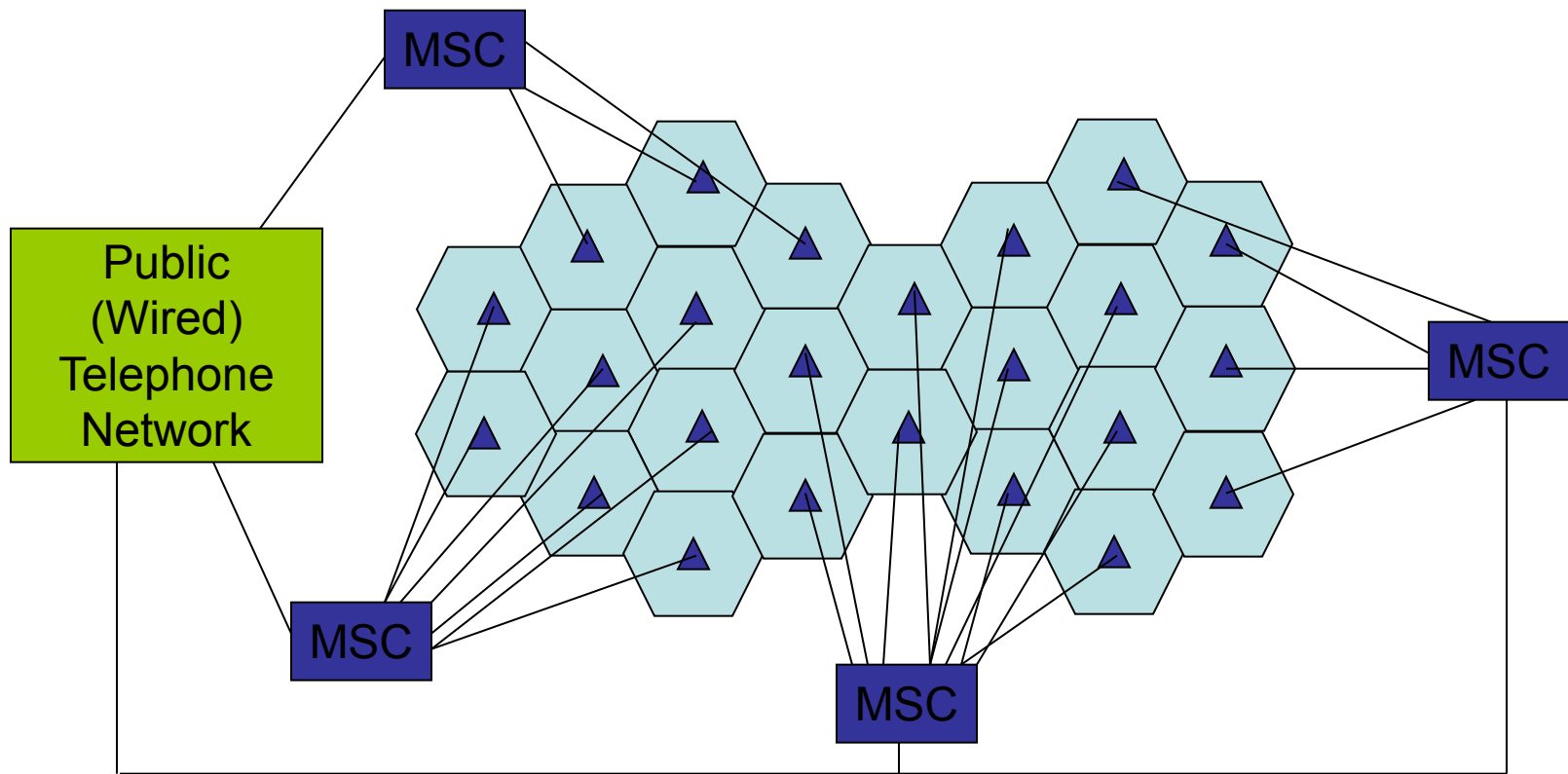
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Wire Main Distribution Frame in Mobile switching Center

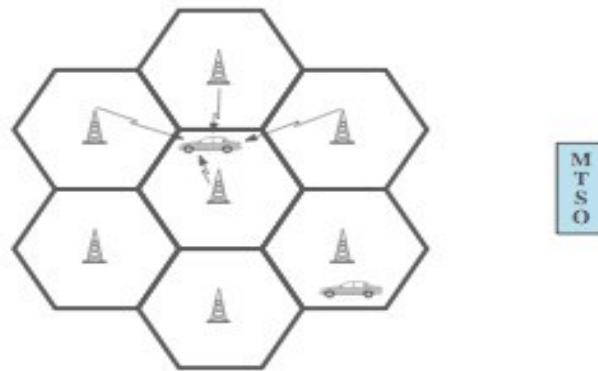


- Mobile switching center (MSC)
 - Coordinates the activities of all the base stations
 - Connect the entire cellular system to the PSTN
 - Accommodates all billing and system maintenance functions

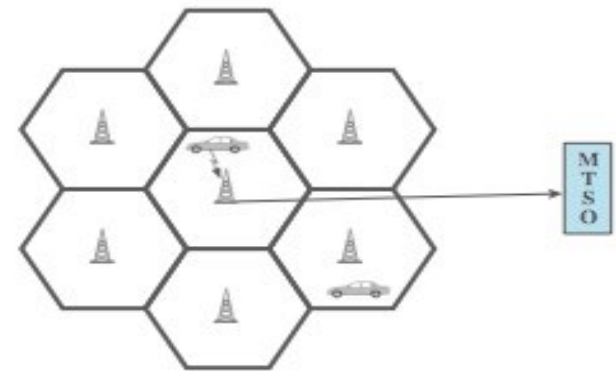
A group of local base stations are connected (may be wire) to a mobile switching center (MSC). MSC is connected to the rest of the world (normal telephone system) or to other MSCs (by wires).



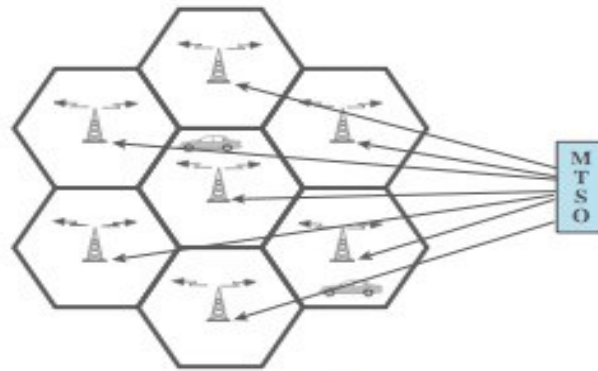
Call Stages



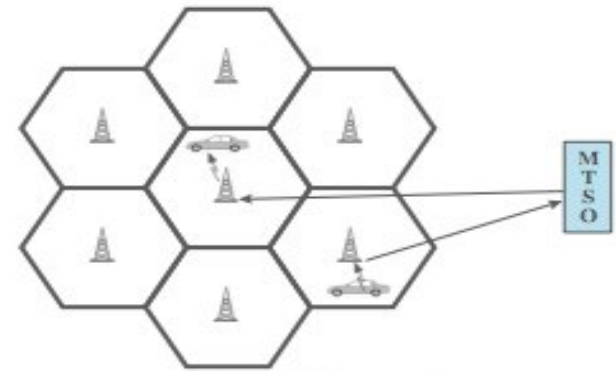
(a) Monitor for strongest signal



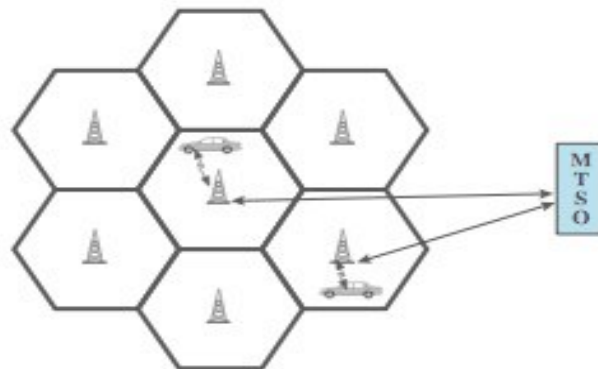
(b) Request for connection



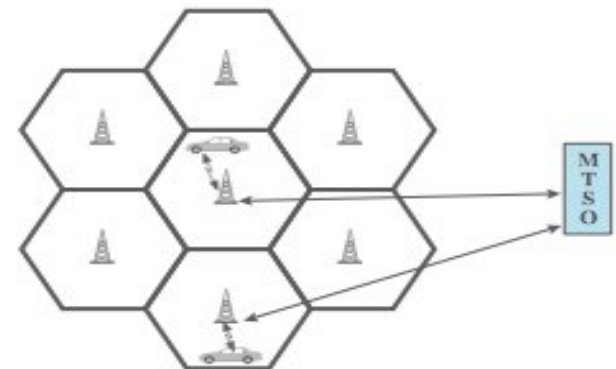
(c) Paging



(d) Call accepted



(e) Ongoing call



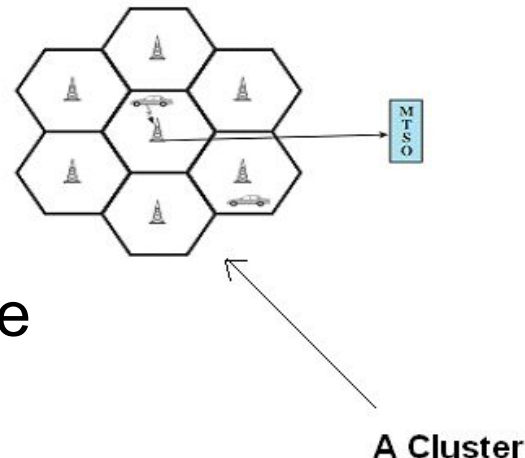
(f) Handoff

Communication between the base station and the mobiles is defined by a standard *common air interface* (CAI) that specifies four different channels. The channels used for voice transmission from the base station to mobiles are called *forward voice channels (FVC)* and the channels used for voice transmission from mobiles to the base station are called *reverse voice channels (RVC)*. The two channels responsible for initiating mobile calls are the *forward control channels (FCC)* and *reverse control channels (RCC)*. Control channels are often called *setup channels* because they are only involved in setting up a call and moving it to an unused voice channel. Control channels transmit and receive data messages that carry call initiation and service requests, and are monitored by mobiles when they do not have a call in progress. Forward control channels also serve as beacons which continually broadcast all of the traffic requests for all mobiles in the system.

Cluster

- Each MSC coordinates a number of base stations
 - The set of base stations controller by a single MSC is called a CLUSTER
 - The number of base stations in a cluster is usually denoted by the letter N

- In AMPS the number of cells inside a cluster is 7
- On the other hand in GSM there are 3 or 4 cells inside a cluster



- Old communication systems use a single high power transmitter and the coverage area is very large. The next base station was so far away that the interference was not an issue.
- However, old systems support just a few users

The cellular concept is a system level idea which calls for replacing a single, high power transmitter (large cell) with many low power transmitters (small cells), each providing coverage to only a small portion of the service area. Each base station is allocated a portion of the total number of channels available to the entire system, and nearby base stations are assigned different groups of channels so that all the available channels are assigned to a relatively small number of neighboring base stations. Neighboring base stations are assigned different groups of channels so that the interference between base stations (and the mobile users under their control) is minimized. By systematically spacing base stations and their channel groups throughout a market, the available channels are distributed throughout the geographic region and may be reused as many times as necessary, so long as the interference between co-channel stations is kept below acceptable levels.

2.2 Frequency Reuse

Cellular radio systems rely on an intelligent allocation and reuse of channels throughout a coverage region [Oet83]. Each cellular base station is allocated a group of radio channels to be used within a small geographic area called a *cell*. Base stations in adjacent cells are assigned channel groups which contain completely different channels than neighboring cells. '

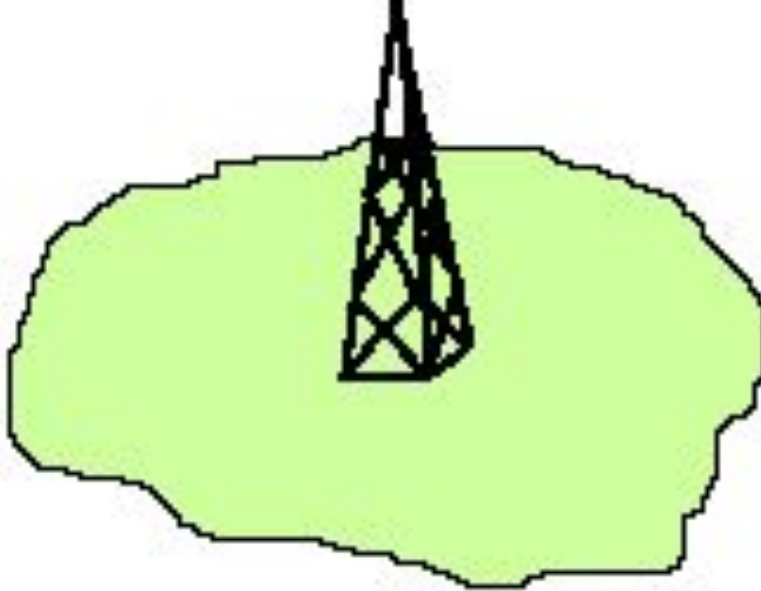
The design process of selecting and allocating channel groups for all of the cellular base stations within a system is called *frequency reuse* or *frequency planning*

Frequency reuse

- is a method used by service providers to improve the efficiency of a cellular network and to serve millions of subscribers using a limited radio spectrum
- is based on the fact that after a distance a radio wave gets attenuated and the signal falls below a point where it can no longer be used or cause any interference
- a transmitter transmitting in a specific frequency range will have only a limited coverage area
- beyond this coverage area, that frequency can be reused by another transmitter

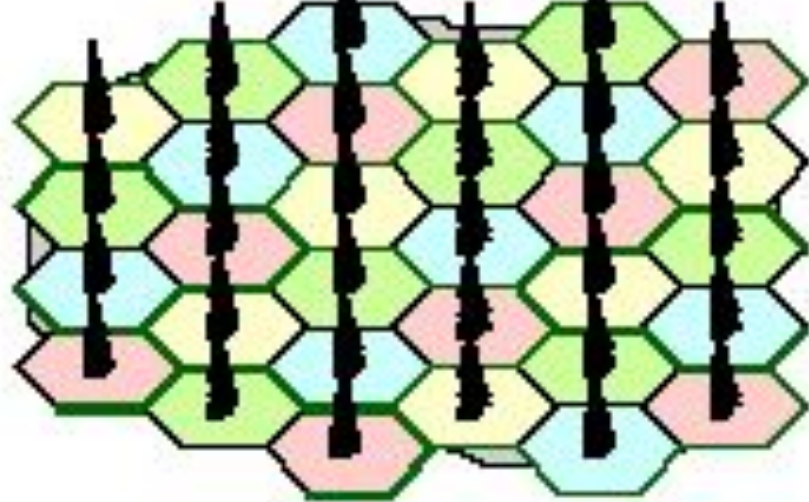
Cellular Networks

OLD radio systems



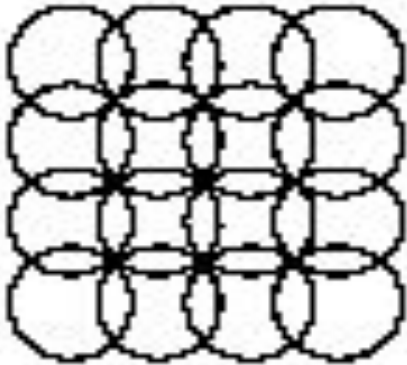
- high-power transmitter
- large coverage area

NEW (Cellular systems)

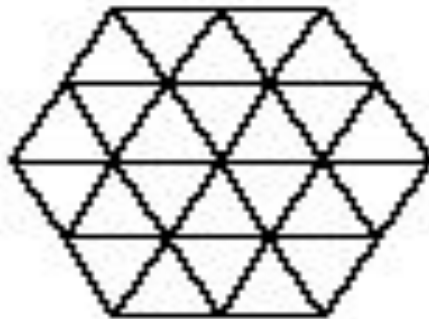


- low-power transmitters
- small coverage areas (cells)
- frequency reuse
- handoff and central control
- cell splitting to increase call capacity

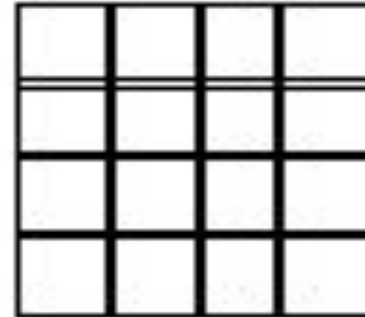
Coverage Patterns



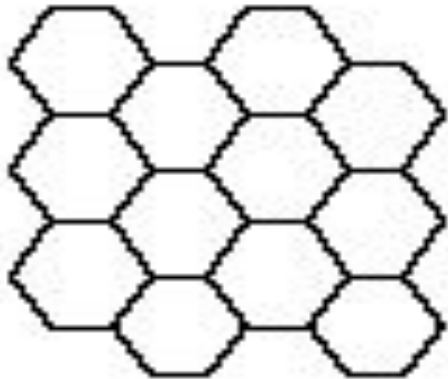
circles



equilateral triangles



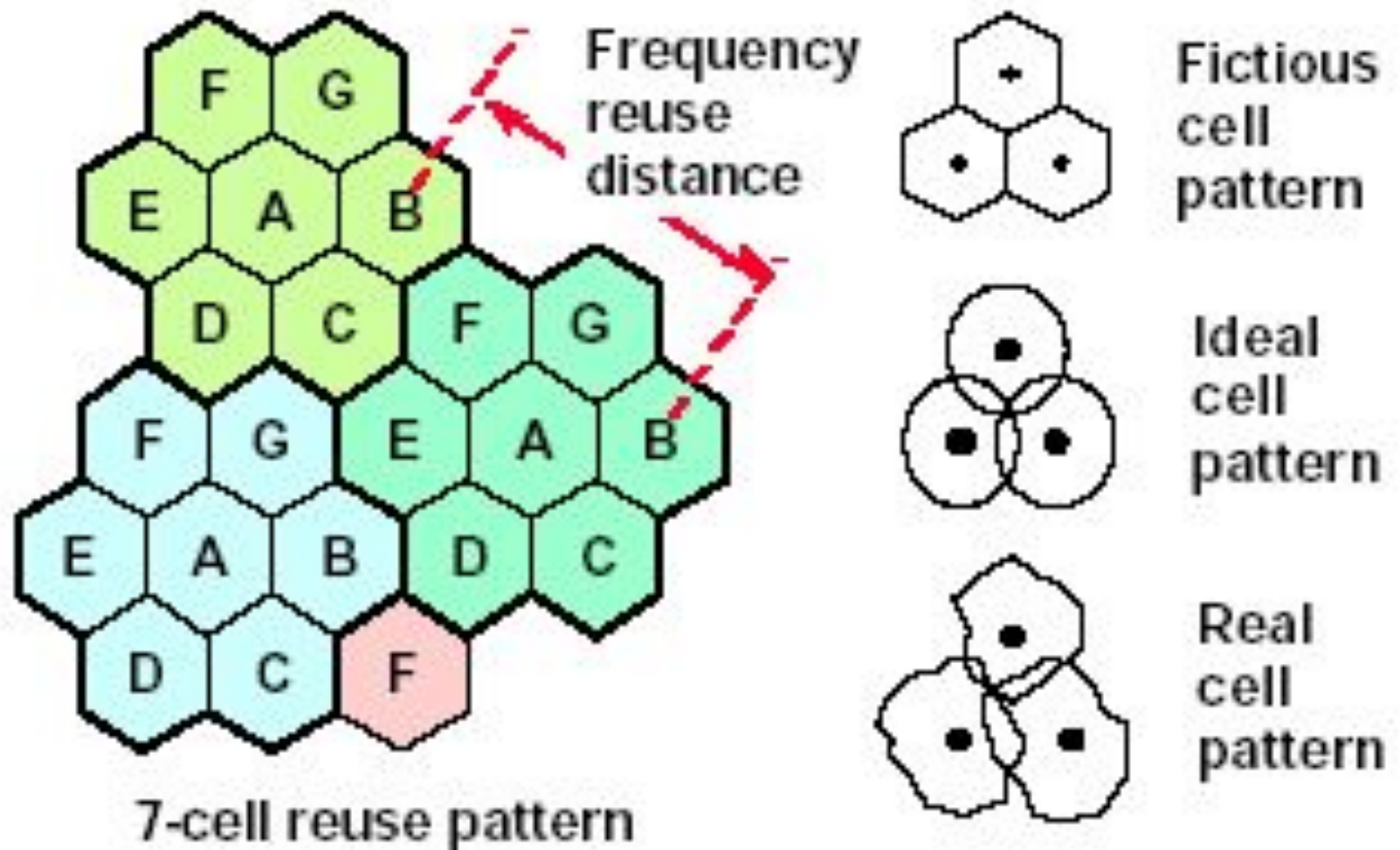
squares



hexagons

geometric shapes which cover an entire region without overlap and with equal area

Cellular Coverage Representation



- Hexagonal cell shape has been universally adopted, since it permits easy and manageable analysis of a cellular system.
- The actual radio coverage of a cell is determined from field measurements or propagation prediction models.

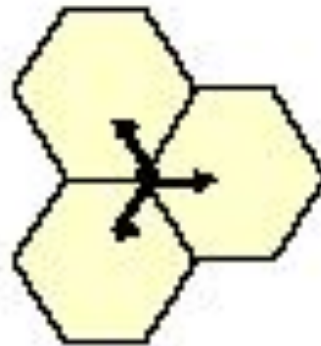
- For a given distance between the center of a polygon and its farthest perimeter points, the hexagon has the largest area among the sensible geometric cell shapes.
- Thus, by using the hexagon geometry, the fewest number of cells can cover a geographic region, and the hexagon also closely approximates a circular radiation pattern which would occur for an omni-directional base station antenna and free space propagation.

- When using hexagons to model coverage areas, base station transmitters are depicted as either being
 - In the center of the cell, or
 - On three of the six cell vertices.
- Normally
 - Omni-directional antennas are used in center-excited cells
 - Sectorized directional antennas are used in corner-excited cells.
 - Practical considerations usually do not allow base stations to be placed exactly as they appear in the hexagonal layout. Most system design permit a base station to be positioned up to one-fourth the cell radius from the ideal location.

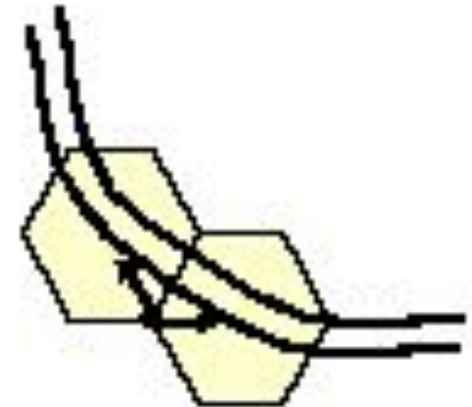
Site Configurations



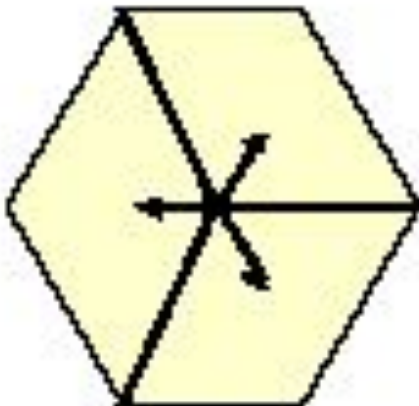
omni-cell



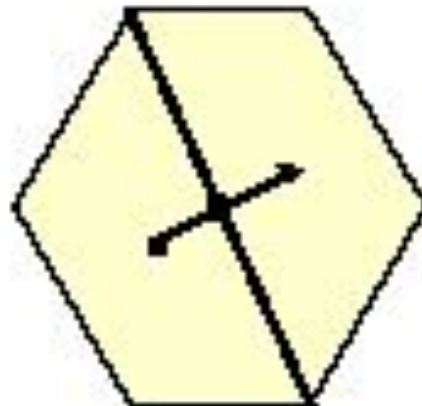
3-sector cloverleaf



2-sector



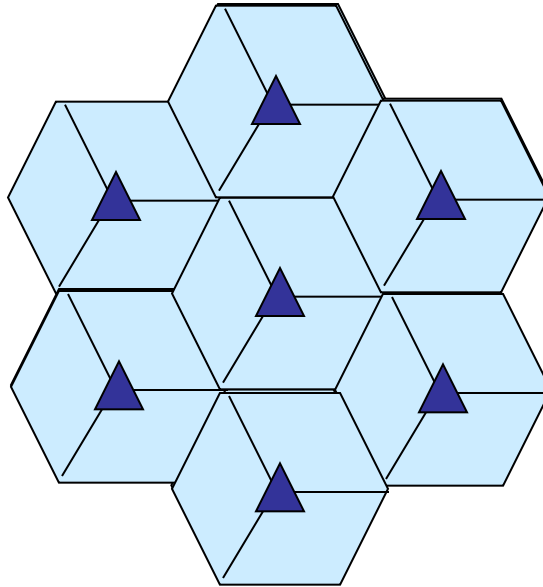
3-sector sectorized cell



2-sector sectorized cell

Directional Antenna at Base Station

With 120 degree antenna, we draw the cells as:



120 Degree Antenna Towers



Coverage map example

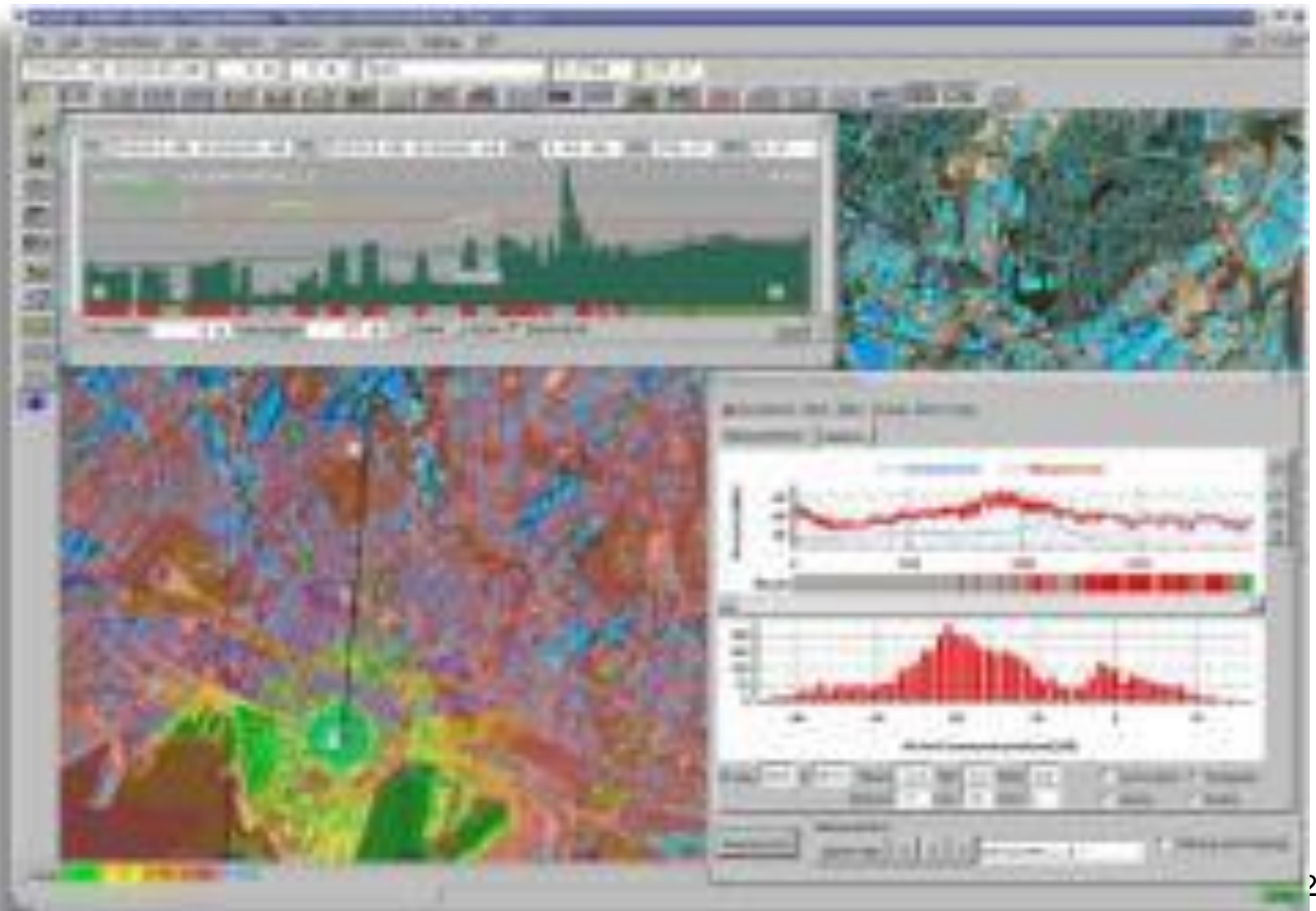
- Unfortunately cell coverage is normally neither hexagonal or circular
- Figure shows coverage example from a city centre
- Complicates radio planning



Radio planning tools

- Radio planning is most often performed assisted by an automated process using a computer
- Underlying functionality
 - Digital maps
 - Propagation modelling
 - System parameters and system performance
 - Traffic assumptions and theory
- Often theoretical computer based modelling can be tuned by real life data
 - Propagation measurements
 - Live network traffic data

Example Tool – Astrix



2 ASTRIX RADIO PLANNING PRODUCT FEATURES

2.1 INTRODUCTION

ASTRIX has all the functionality of an advanced radio planning tool. GSM functionality is an integrated part of the ASTRIX 2G module and allows simultaneous view of e.g. GSM900/1800 networks and UMTS networks. This allows the user to plan mixed-mode networks, to have data consistency across the networks and to plan with site re-use.

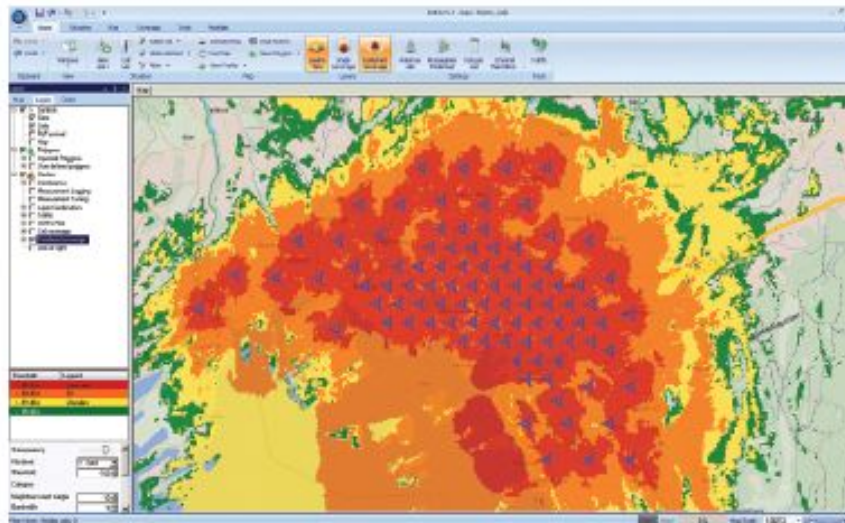


Figure 1 ASTRIX main window

2.2 SEAMLESS MAP HANDLING

ASTRIX uses an integrated GIS (Geographical Information System) as the basis for map handling. The GIS system provides advanced map features for best possible views and management of maps. The GIS system supports dynamic map resolution and various

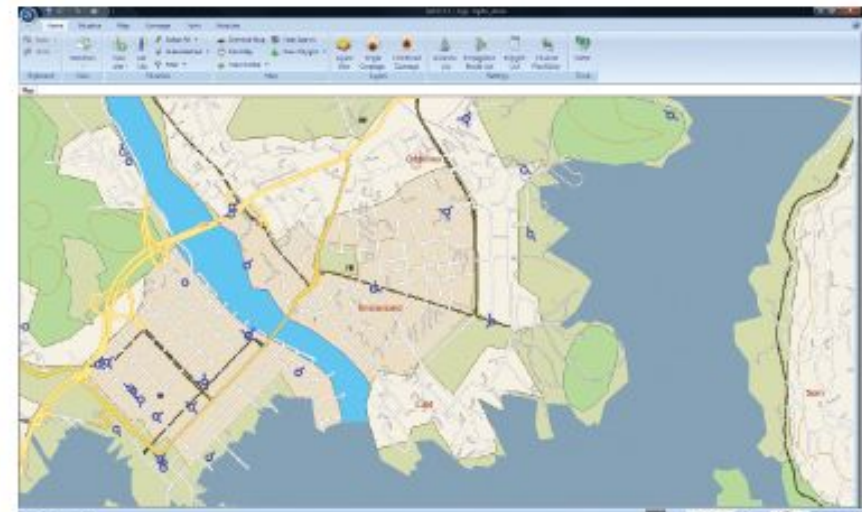
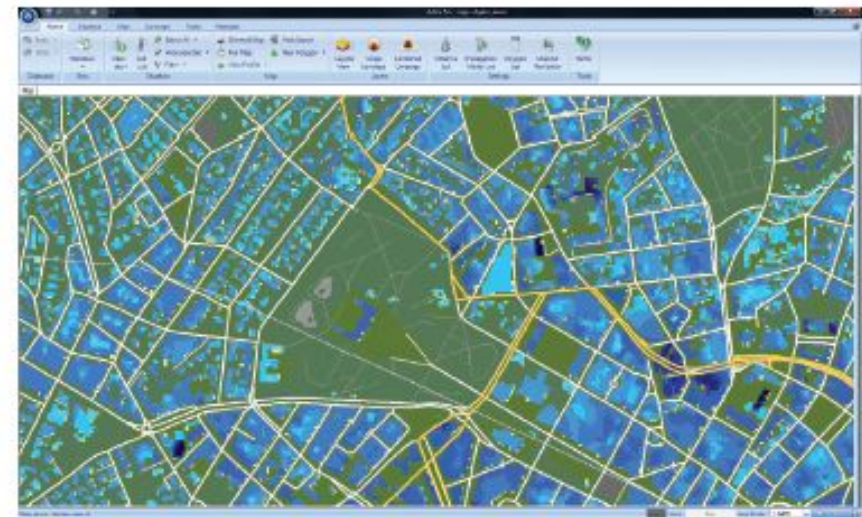


Figure 2 Examples of ASTRIX map layers

The following map layers are supported:

- Raster/grid ground heights
- Raster/grid morphology
- Vector/linear features

There are other cell design tools



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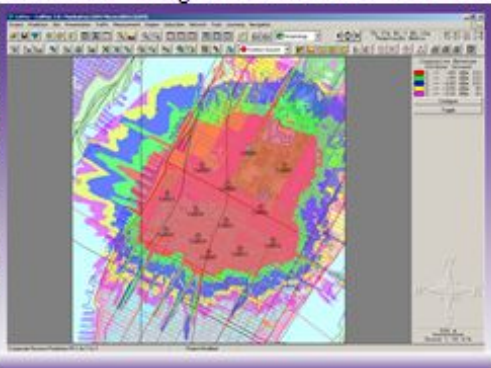
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CelPlanner[™]

CelPlanner[™] is one of the main components of the industry acclaimed *CelPlanner[™]* Suite, the most complete and integrated suite of software tools addressing the complexity of Wireless Networks design from conception to implementation.

CelPlan[®] offers *CelPlanner[™]* as an advanced integrated software tool that addresses today's and tomorrow's Wireless Network Planning and Design needs by supporting wireless access technologies such as 4G (WiFi, Wimax, DVB-H), 3G (IS-2000, 1xRTT, UMTS), 2.5G (GPRS/EDGE), 2G (cdmaOne, IS-95, IS-136, TDMA, GSM), and analog (AMPS, ETACS), thus allowing seamless and smooth network planning evolution for wireless operators and associates.



CelPlan[®]'s new Wi-Fi/ WiMax Module fully implements OFDM link adaptation. The module supports systems such as 802.11a/b, 802.16-OFDM, and 802.16-OFDMA.

Amongst the many features that support the design of such networks, users can take advantage of the availability of multiple channel bandwidths and detailed traffic simulation processes. The whole network designed is based on user-defined radio,

Multi-Technologies

cdmaOne/IS-95

DVB-H

Fixed Wireless

GSM

IS-2000/1xRTT/1xEVDO

Microwave

Paging

TDMA/IS-136

UMTS

UWB

WiFi - WiMax

Cell Planning

- k = the number of channels allocated to each cell in a cluster
- N = cluster size (number of cells in a cluster)
- M = number of clusters within a communication system
- The number of channels available in a cluster is

$$S = kN$$

- The capacity of the cellular systems is

$$C = MS \text{ which is } C = MkN$$

- The frequency reuse factor is $1/N$

Frequency reuse

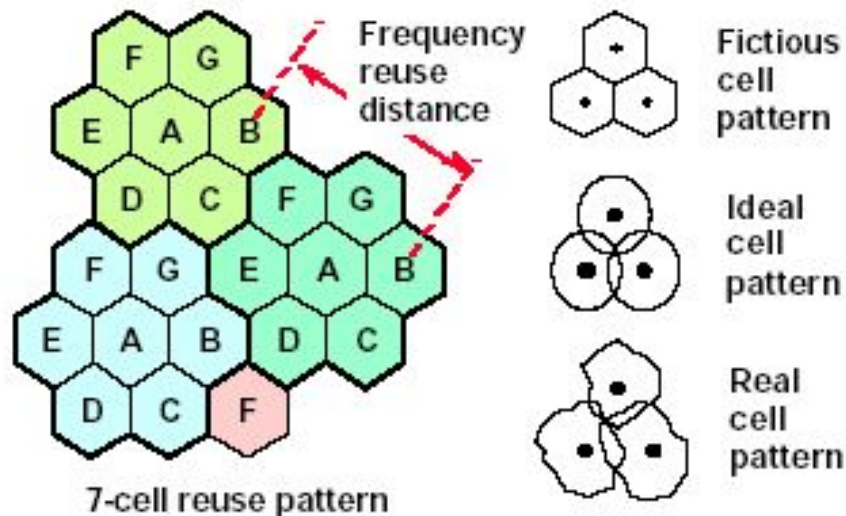
- is a method used by service providers to improve the efficiency of a cellular network and to serve millions of subscribers using a limited radio spectrum
- is based on the fact that after a distance a radio wave gets attenuated and the signal falls below a point where it can no longer be used or cause any interference
- a transmitter transmitting in a specific frequency range will have only a limited coverage area
- beyond this coverage area, that frequency can be reused by another transmitter

- In order to tessellate (mozaikle dosemek) - to connect without gaps between adjacent cells – the geometry of the hexagons is such that the number of cells per cluster, N , can only have values

$$N = i^2 + ij + j^2$$

where i and j are non-negative integers, i.e.,
 $i \geq 0, j \geq 0$

- The factor N is typically equal to 4, 7, 12,



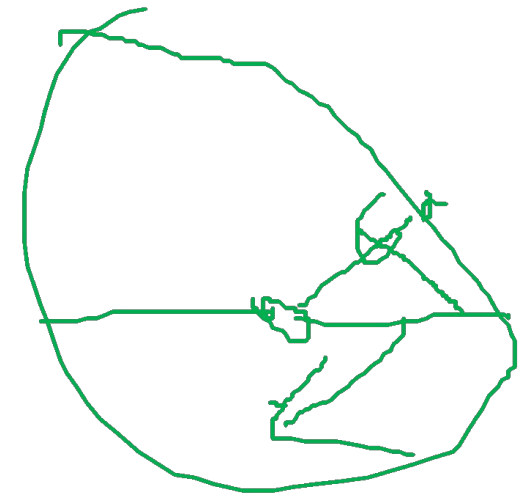
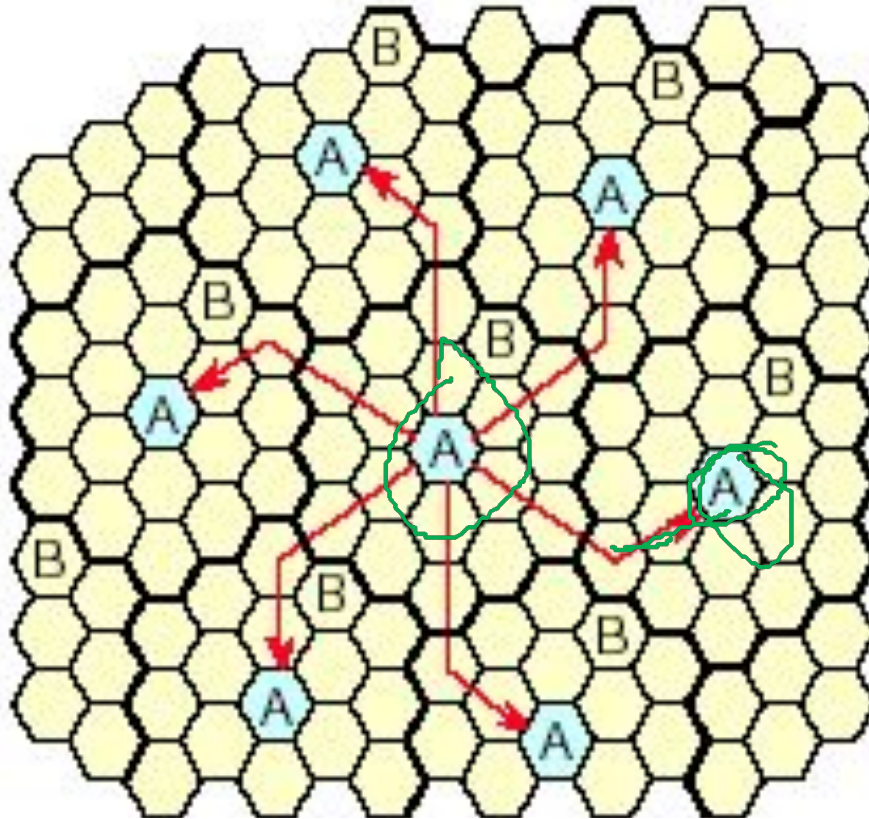
Co-channel cells

- Frequency reuse implies that in a given coverage area there are several cells that use the same set of frequencies. These cells are called c-channel cells, and the interference between signals from these cells is called co-channel interference.

To find the nearest co-channel cell

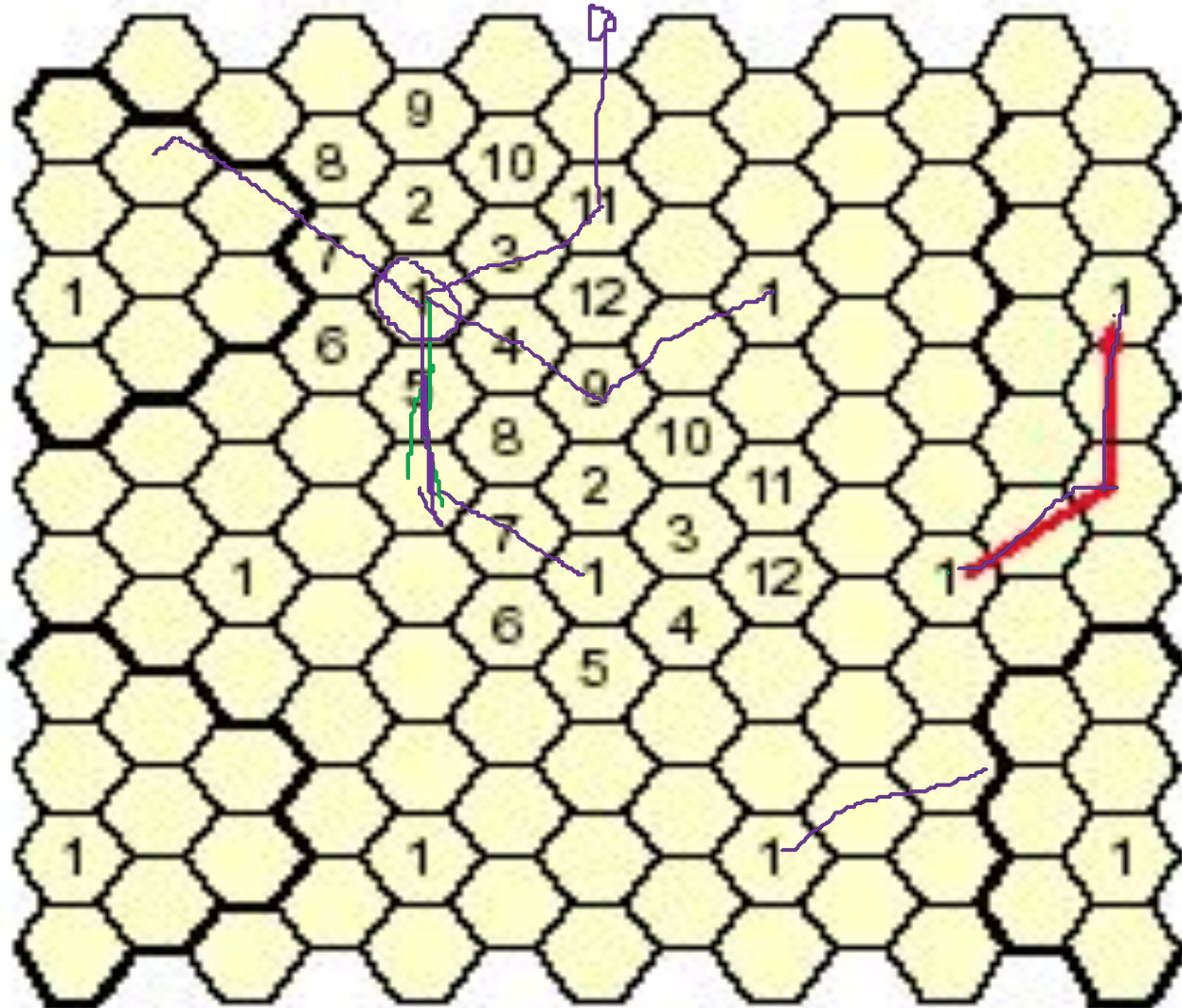
To find the nearest co-channel neighbors of a particular cell, one must do the following: (1) move i cells along any chain of hexagons and then (2) turn 60 degrees counter-clockwise and move j cells.

19-cell reuse pattern ($i=3, j=2$)

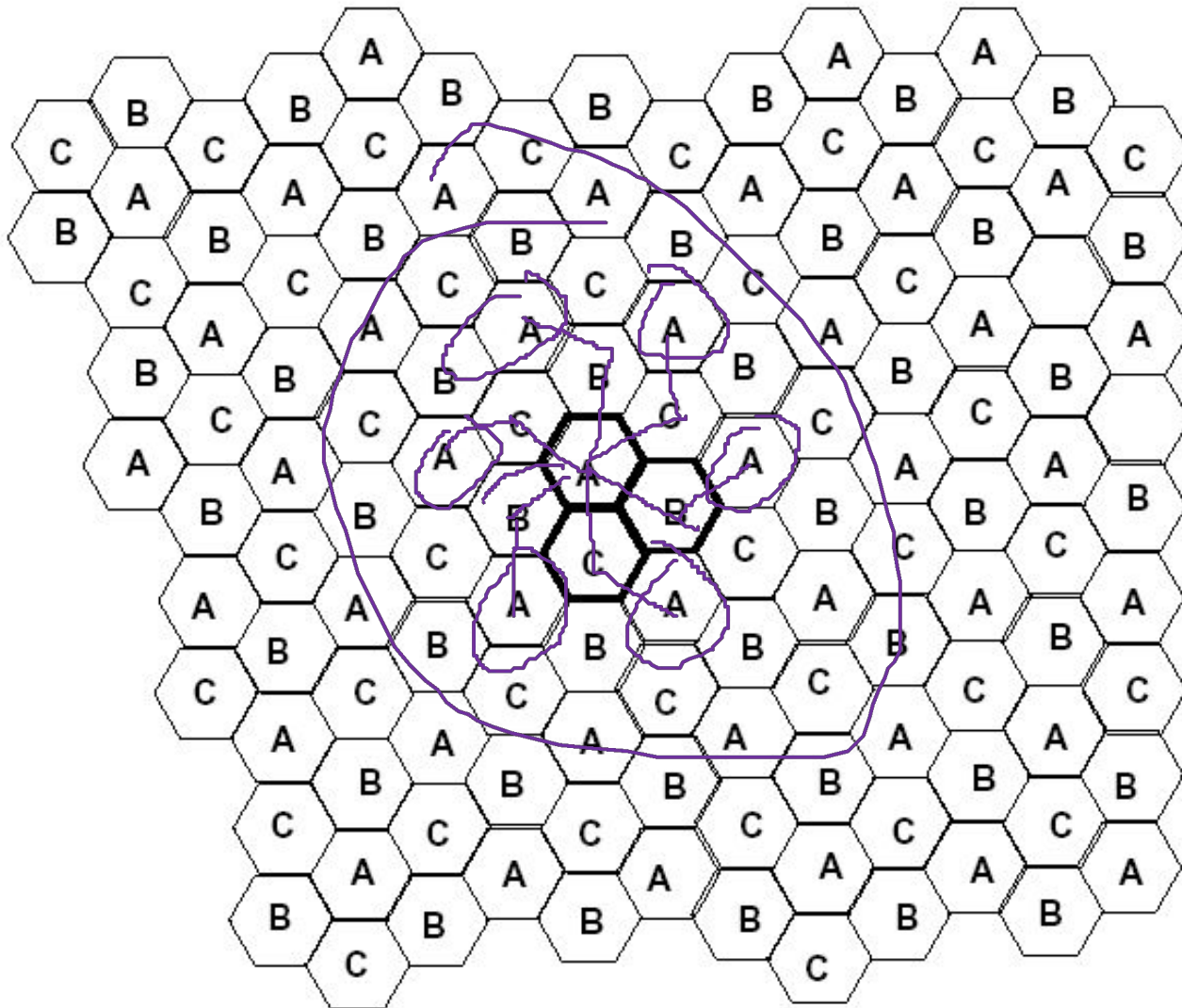


To find the nearest co-channel neighbors of a particular cell, one must do the following: (1) move i cells along any chain of hexagons and then (2) turn 60 degrees counter-clockwise and move j cells.

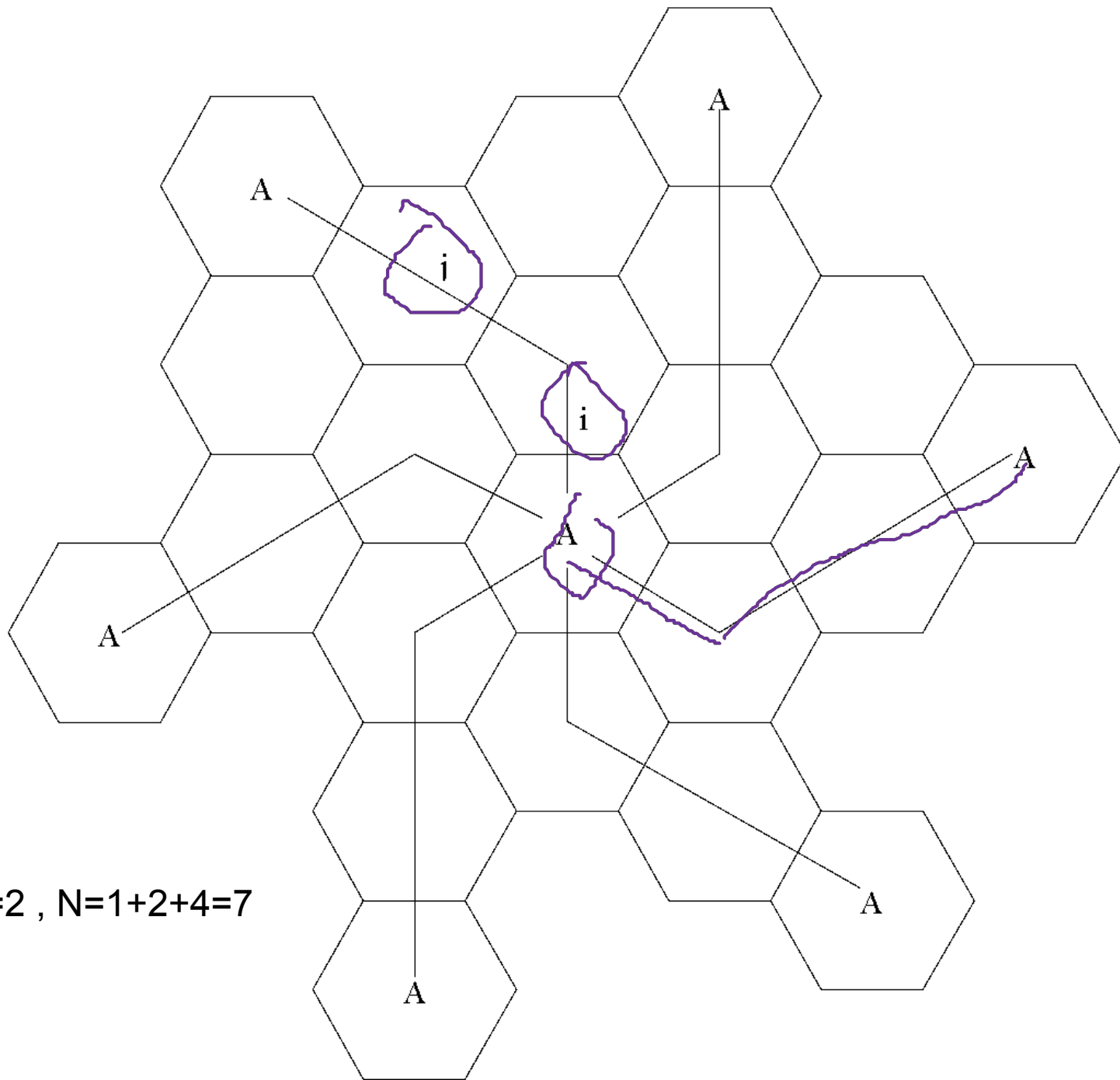
12-cell reuse pattern ($i=2, j=2$)



3-cell reuse pattern ($i=1, j=1$)

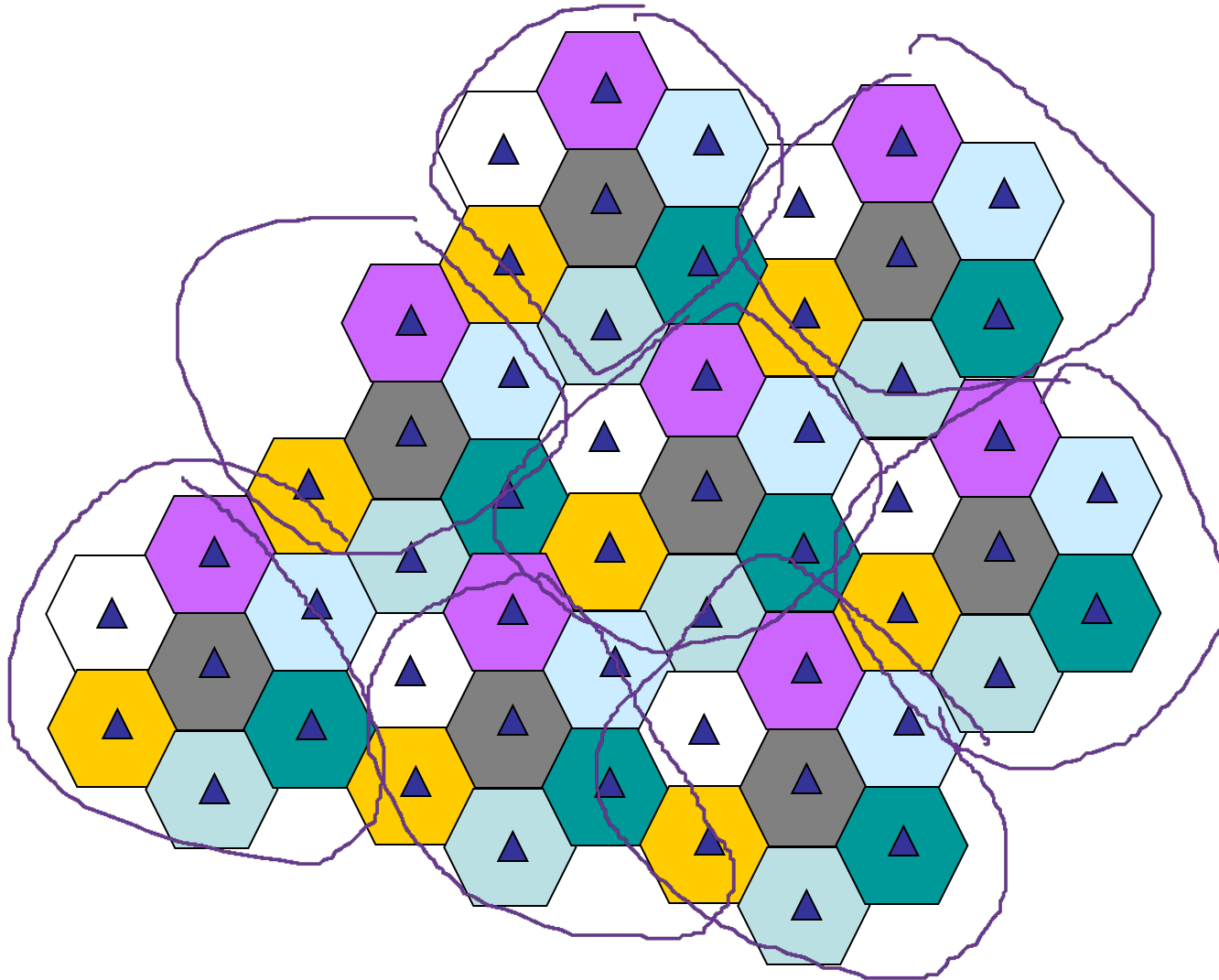


A large hexagonal grid of 100 hexagons arranged in a larger hexagonal shape. Each hexagon contains one of the letters A, B, C, or D. A central cluster of three hexagons (A, B, D) is highlighted with a thick black border.

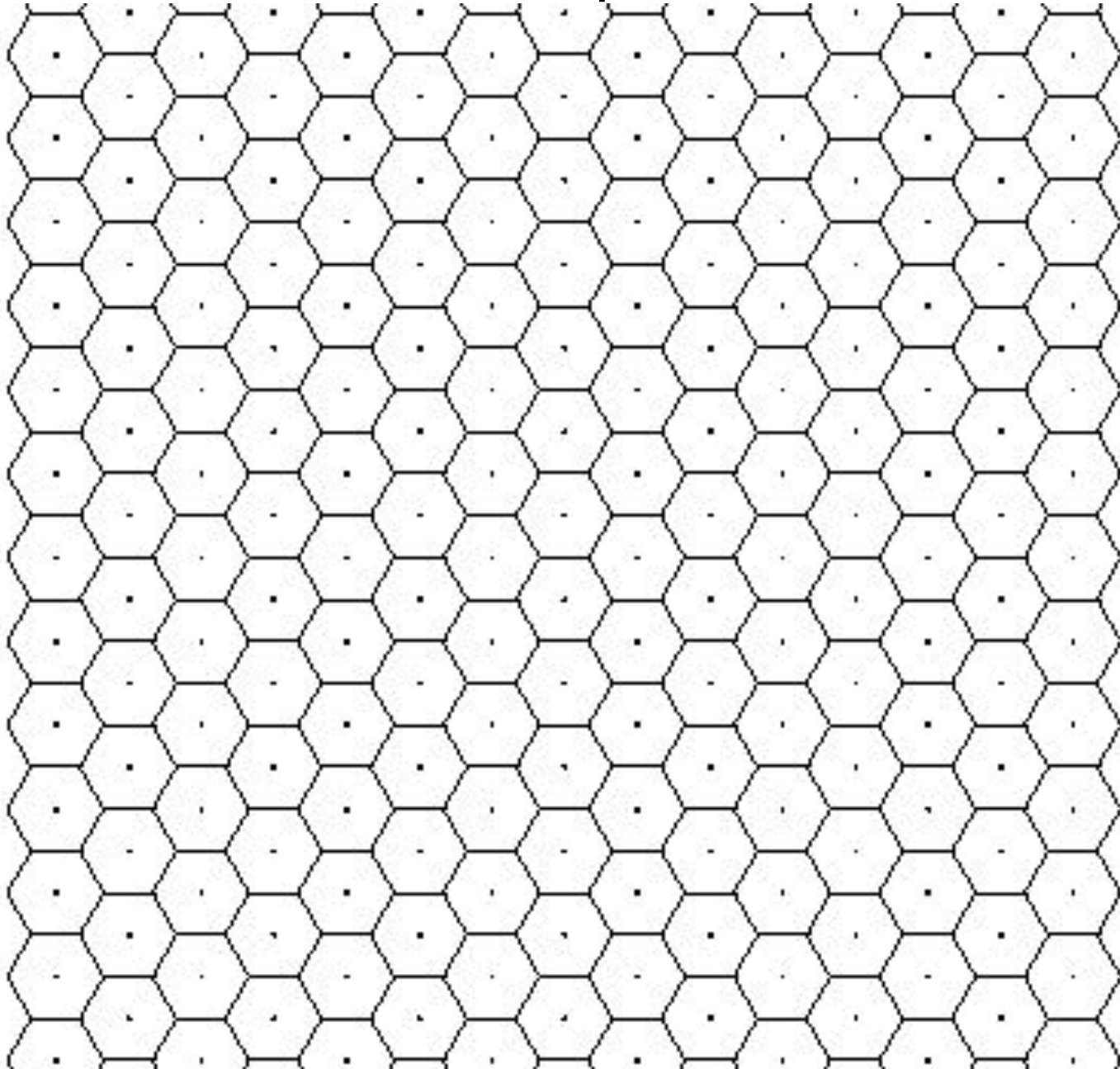


$$i=1, j=2, N=1+2+4=7$$

Cluster size of 7, Reuse Pattern



Exercise: Locate frequencies for $N=3$ or 7



Find the proof of: the number of cells in a cluster equals

$$N=i^2+ij+j^2$$

Example 2.1

If a total of 33 MHz of bandwidth is allocated to a particular FDD cellular telephone system which uses two 25 kHz simplex channels to provide full duplex voice and control channels, compute the number of channels available per cell if a system uses (a) 4-cell reuse, (b) 7-cell reuse (c) 12-cell reuse.

Solution to Example 2.1

Given:

Total bandwidth = 33 MHz

Channel bandwidth = 25 kHz $\times$ 2 simplex channels = 50 kHz/duplex channel

Total available channels = 33,000/50 = 660 channels

(a) For $N = 4$,

total number of channels available per cell = $660/4 \approx 165$ channels.

(b) For $N = 7$,

total number of channels available per cell = $660/7 \approx 95$ channels.

(c) For $N = 12$,

total number of channels available per cell = $660/12 \approx 55$ channels.

Example

If a total of 55 MHz of bandwidth is allocated to a particular FDD cellular telephone system which uses two 50 kHz simplex channels to provide full duplex channels. Compute the cluster size if the number of duplex channels per cell is 25 .

What should be cluster size (N ?)

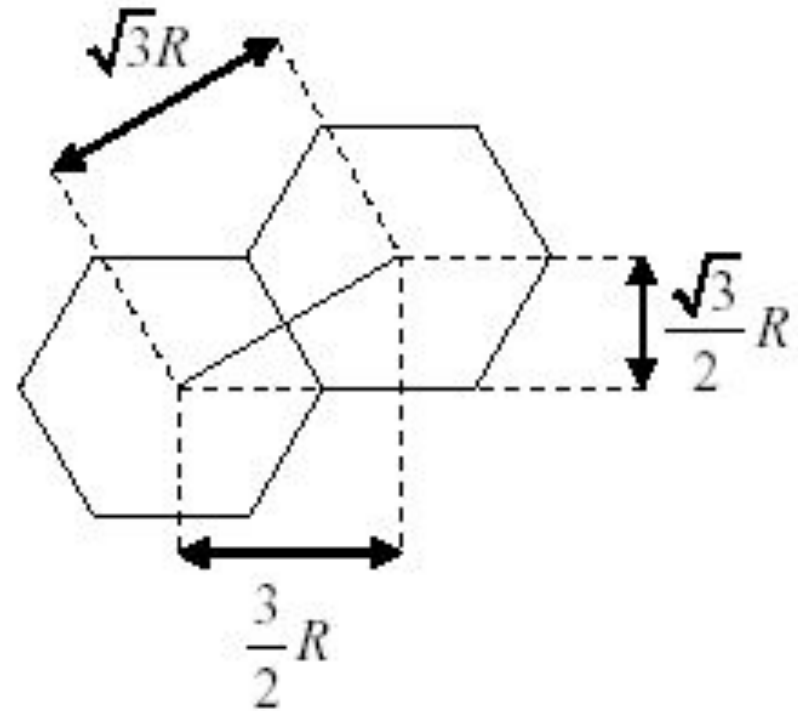
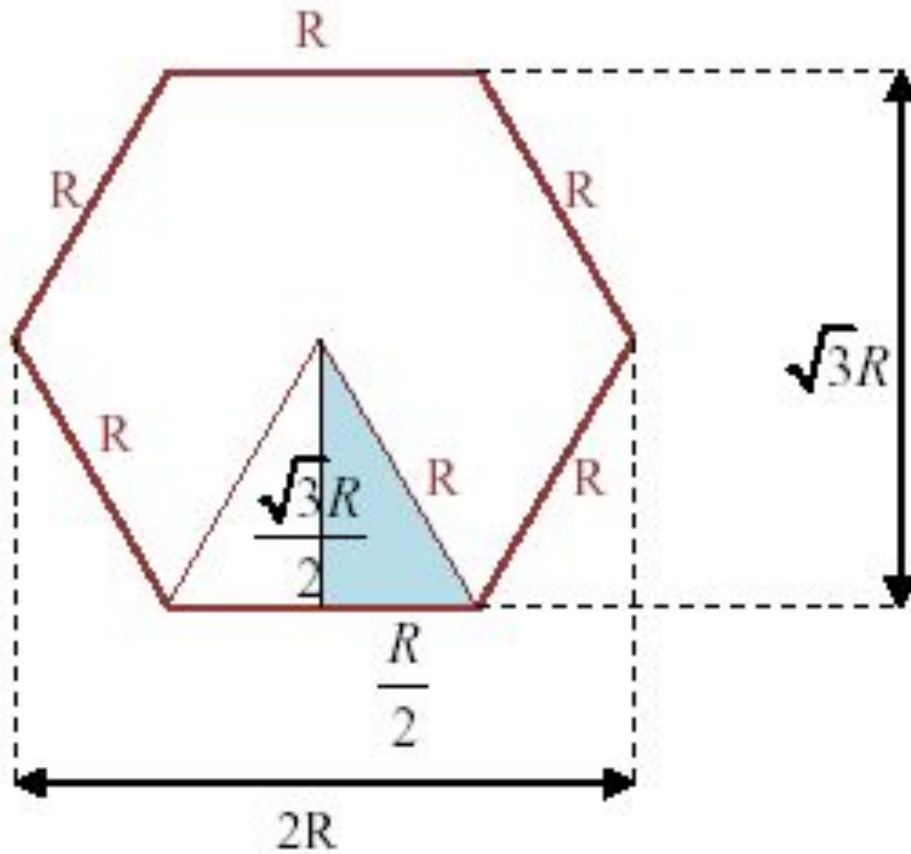
the capacity of a cellular system is directly proportional to the number of times a cluster is replicated in a fixed service area. The factor N is called the *cluster size* and is typically equal to 4, 7, or 12. If the cluster size N is reduced while the cell size is kept constant, more clusters are required to cover a given area and hence more capacity (a larger value of C) is achieved. A large cluster size indicates that the ratio between the cell radius and the distance between co-channel cells is large. Conversely, a small cluster size indicates that co-channel cells are located much closer together. The value for N is a function of how much interference a mobile or base station can tolerate while maintaining a sufficient quality of communications. From a design viewpoint, the smallest possible value of N is desirable in order to maximize capacity over a given coverage area

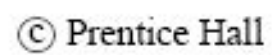
Fundamentals

- Planning and deploying a GSM network is from an operator's point of view a question of:
 - Build as few sites as possible, while maintaining required coverage and capacity
 - Trade off

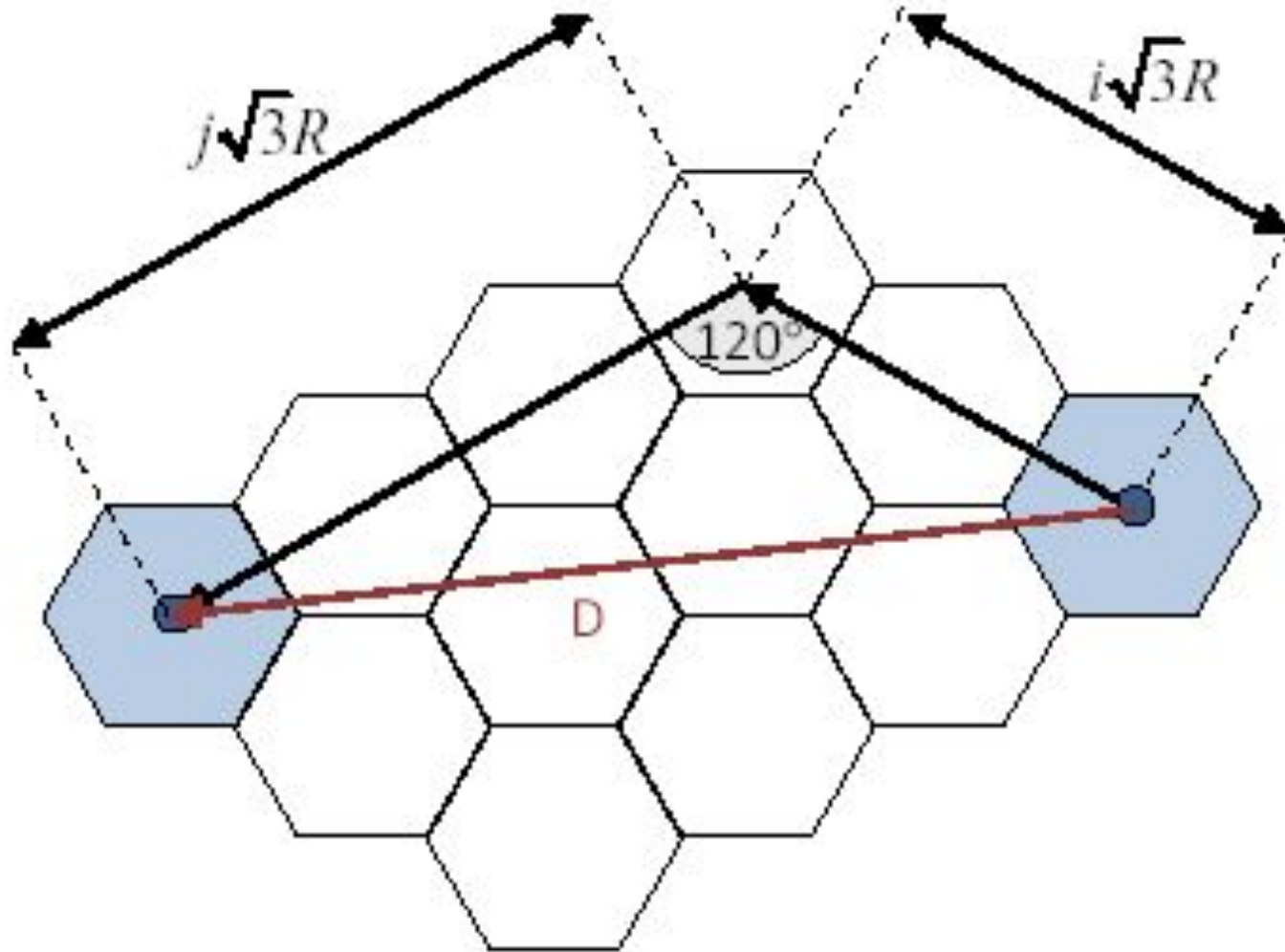


Hexagon Geometry





Finding the distance between co-channels



$$\begin{aligned}
 D^2 &= (i \cdot \sqrt{3}R)^2 + (j \cdot \sqrt{3}R)^2 - 2(i \cdot \sqrt{3}R)(j \cdot \sqrt{3}R) \cos 120^\circ \\
 &= (i^2 + j^2 + ij)2R^2
 \end{aligned}$$

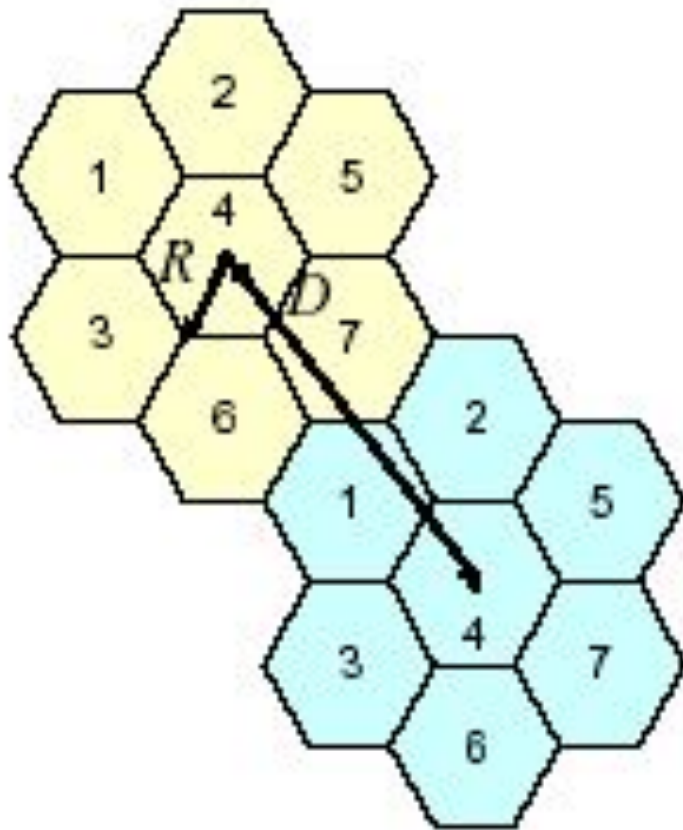
It can be shown that the cell cluster size is given by

$$N = i^2 + j^2 + ij.$$

$\Rightarrow$

$$D^2 = 3NR^2, \quad \text{or} \quad D = \sqrt{3N}R.$$

Relationship between Q and N



Q = co-channel reuse ratio

D = frequency reuse distance

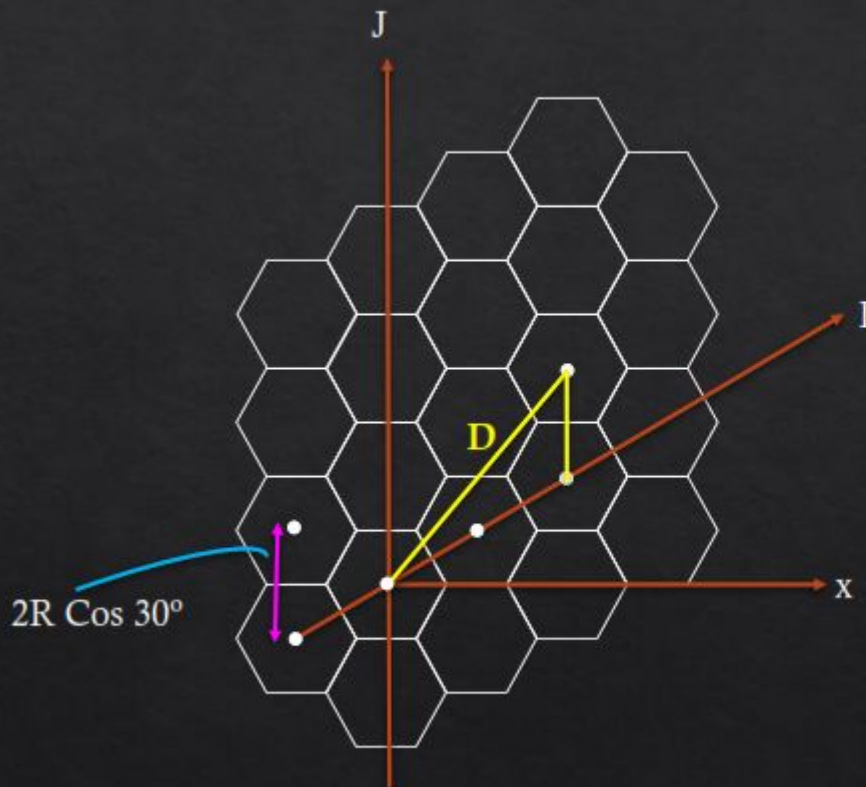
R = cell radius

N = # cells per cluster

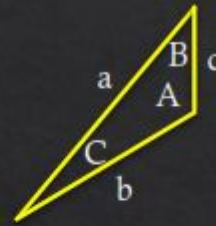
$1/N$ = frequency reuse factor

$$Q = \frac{D}{R} = \sqrt{3N}$$

Proof of Co-channel Frequency Reuse Ratio



$$N = i^2 + i j + j^2 \text{ ----- (1)}$$



Cosine Rule

$$a^2 = b^2 + c^2 - 2bc \cos(A)$$

$$d = 2 R \cos 30^\circ = 2 R \frac{\sqrt{3}}{2} = \sqrt{3} R \quad \text{----- (2)}$$

$$D^2 = (i \times d)^2 + (j \times d)^2 - 2(i \times d) \times (j \times d) \cos 120^\circ$$

$$D^2 = i^2 d^2 + j^2 d^2 - 2 i j d^2 \cos 120^\circ$$

$$D^2 = d^2 \left(i^2 + j^2 - 2ij \left(\frac{-1}{2} \right) \right)$$

$$D^2 = d^2 (i^2 + j^2 + k^2)$$

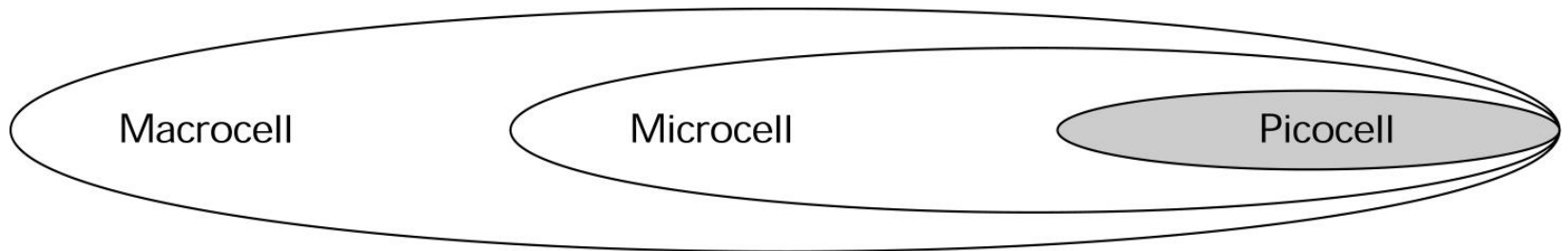
$$D^2 = 3 R^2 (i^2 + i j + j^2) \text{ (Substituting equation 2)}$$

$$D^2 = 3 R^2 N$$

$$q = \frac{D}{R} = \sqrt{3N}$$

Hierarchical cell structures

- In a GSM system it is common that cells of different sizes co-exist in that same area:
 - Picocells, microcells, macrocells
- This is called *hierarchical cell structure*
- Can make handover (cell change) complicated. Often different types of users are reserved for one cell type, e.g.:



Mixed Cell Architecture

